MOBL-07: Network Request Monitoring

> **Series:** MOBL — Mobile Monitoring | **Notebook:** 7 of 12 | **Created:** February 2026 | **Last Updated:** 10/29/2026

Overview

Every mobile app depends on the network. Whether it is fetching user profiles, loading product catalogs, or submitting orders, HTTP(S) requests are the lifeline between a mobile frontend and its backend services. Dynatrace automatically captures these network requests from instrumented mobile apps, providing deep visibility into:

- **What** is being requested (URL, method, status code)
- **How long** each request takes (`duration`)
- **How much** data is transferred (request and response sizes)
- **What connection** the device is using (WiFi, 5G, LTE, 3G)
- **Where** the request goes on the backend (distributed trace correlation)

This notebook explains how Dynatrace captures network requests across platforms, how to query and analyze them with DQL, and how to identify slow or failed requests that degrade the mobile user experience.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Grail
Mobile App Instrumented	Dynatrace Mobile SDK deployed (iOS, Android, Flutter, or React Native)
Network Requests	Mobile app actively making HTTP(S) requests being captured by the SDK
Permissions	`storage:user.events:read`, `storage:spans:read`
Distributed Tracing	Backend services instrumented with OneAgent, OpenTelemetry, or both for correlation

1. Automatic HTTP Capture

The Dynatrace Mobile SDK automatically intercepts HTTP(S) requests made through standard networking libraries on each platform. No manual instrumentation is required for the libraries listed below.

Supported Platforms and Libraries

Platform	HTTP Library	Auto-Captured
iOS	URLSession	Yes
iOS	Alamofire (URLSession-based)	Yes
Android	OkHttp	Yes
Android	HttpURLConnection	Yes

Platform	HTTP Library	Auto-Captured
Flutter	Dart `http` package via `Dynatrace().createHttpClient()`	Yes (wrapper client)
Flutter	`dart:io HttpClient` , gRPC, custom stacks	No -- manual web request monitoring
React Native	Fetch API, XMLHttpRequest	Yes

> **Note:** Third-party libraries that use non-standard HTTP stacks (e.g., custom socket implementations) may require manual instrumentation. Check the Dynatrace documentation for your specific SDK version.

Data Captured Per Request

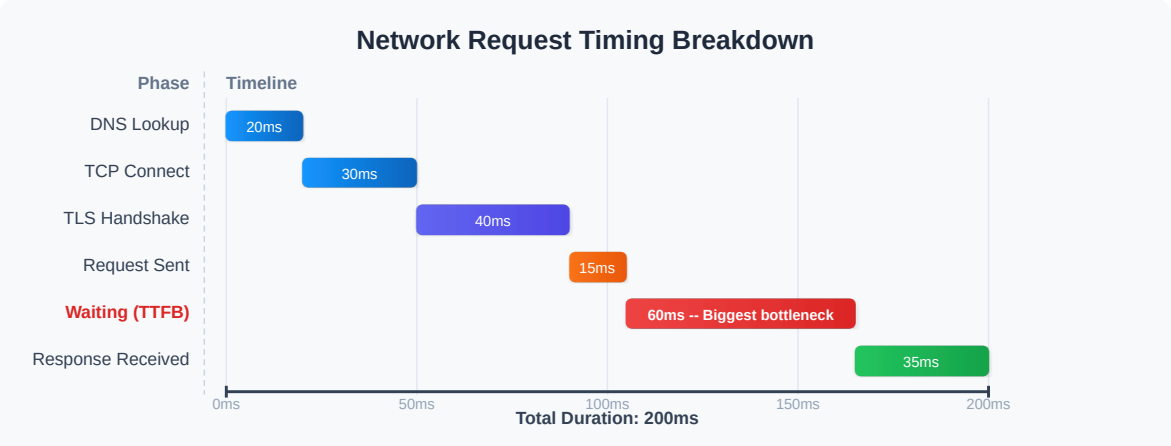
For each intercepted network request, the SDK captures:

Data Point	Description
URL	Full request URL (query parameters may be redacted based on privacy settings)
HTTP Method	GET, POST, PUT, DELETE, PATCH, etc.
Status Code	HTTP response status code (200, 404, 500, etc.)
Request / Response Size	Body sizes in bytes (`http.request.body.size` , `http.response.body.size`) where the agent reports them
Duration	Total time from request initiation to response completion
Connection Type	Network connection type at the time of the request (WiFi, LTE, 5G, etc.)

The SDK batches captured data and sends it to the Dynatrace cluster at regular intervals, minimizing the performance impact on the mobile app.

2. Connection Type Detection

Understanding the network conditions under which requests are made is critical for diagnosing performance issues. A request that takes 3 seconds over a 2G connection is expected, but the same latency over WiFi signals a backend problem.



A diagnostic model of request phases. Mobile request events record the total duration, not each phase.

Connection Types

The SDK records the connection type at the time of each request in `network.connection.type` (dictionary examples `wifi`, `cell`) and the technology in `network.connection.subtype` (examples `lte`, `802.11x`); the cellular generations below are subtype values. Run `summarize count(), by:{network.connection.type, network.connection.subtype}` on your own data to see the exact values your agents report:

Connection	Description
WiFi	Connected via WiFi network
5G	5th generation cellular network
LTE/4G	4th generation cellular network
3G	3rd generation cellular network
2G	2nd generation cellular network

Connection type is stored alongside each network request event, enabling you to:

- **Segment performance by connection type** to set realistic SLOs

- **Identify users on poor connections** who experience degraded performance
- **Build adaptive behavior** in your app based on detected connection quality

3. Request Timing Breakdown

A single network request goes through multiple phases, each of which can contribute to perceived latency. This section is a **diagnostic framework**, not a list of captured fields: a mobile request event reports the overall `duration`, status and sizes. The phase-level timings in the `rum_request` model come from the browser's W3C Resource Timing API (`characteristics.has_w3c_resource_timings`) and apply to web frontends. For a slow mobile request, compare `duration` across connection types (Section 2) and follow the backend trace (Section 4) to separate network time from server time.

> **Dictionary:** model `rum_request` (`user.events`) lists `characteristics.has_w3c_resource_timings`, `characteristics.has_w3c_navigation_timings`, `http.request.method`, `http.response.status_code` — no DNS/TCP/TLS phase fields; `network.connection.type` (`experimental`); `http.response.body.size` (`stable`), read 10/02/2026.

Timing Phases

Phase	What Happens	What a Slow Phase Indicates
DNS Lookup	Resolves the domain name to an IP address	DNS server issues, missing DNS cache, poor network conditions
TCP Connect	Establishes a TCP connection to the server	Network latency, server unreachable, firewall delays
TLS Handshake	Negotiates the TLS/SSL secure connection	Certificate chain issues, slow TLS negotiation, outdated cipher suites
Request Sent	Transmits the HTTP request body to the server	Large request payload, low upload bandwidth
Waiting (TTFB)	Time to First Byte -- server processes the request	Slow backend logic, database queries, cold starts, overloaded services

Phase	What Happens	What a Slow Phase Indicates
Response Received	Downloads the HTTP response body	Large response payload, low download bandwidth, connection throttling

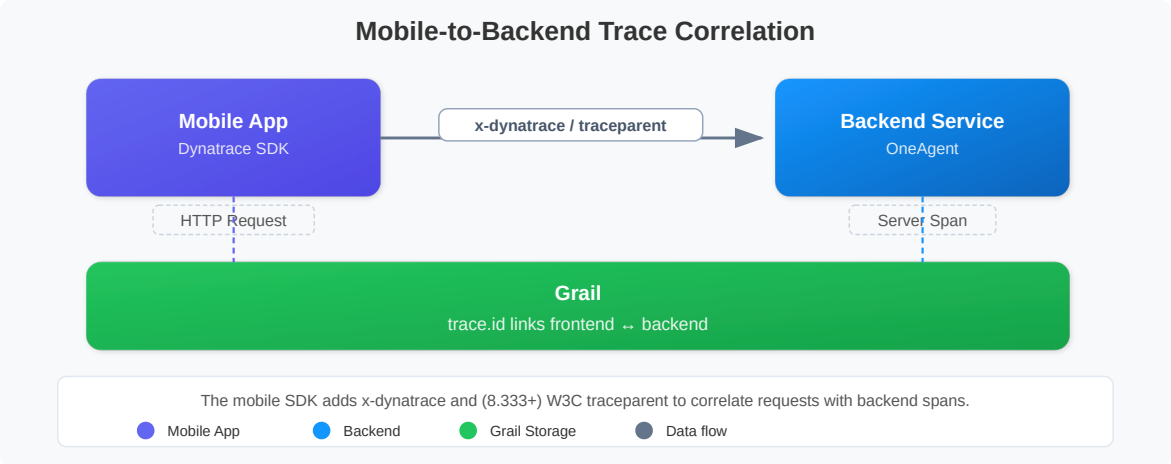
Diagnostic Guide

Symptom	Likely Cause	Investigation
High DNS time	DNS resolution issues	Check DNS provider, enable DNS caching
High TCP Connect time	Network path issues	Check CDN proximity, network hops
High TLS time	Certificate or cipher issues	Verify certificate chain, enable TLS session resumption
High TTFB	Backend performance	Check backend traces, database queries, cold starts
High Response time	Large payloads	Enable compression, paginate responses, reduce payload size

> **Tip:** When the backend span accounts for most of the request's `duration`, the problem is on the backend. Use the frontend-to-backend correlation (Section 4) to trace the request into your server-side services.

4. Frontend-to-Backend Correlation

One of the most powerful features of Dynatrace mobile monitoring is the ability to correlate a network request on the mobile device with the corresponding backend service trace. This creates end-to-end visibility from the user's finger tap to the database query.



How It Works

1. **Request Initiation** -- The mobile app makes an HTTP(S) request using a standard networking library.
2. **Header Injection** -- The Dynatrace Mobile SDK adds the `x-dynatrace` header to the outgoing request (RUM Classic), and from OneAgent for Mobile 8.333 also injects the W3C Trace Context headers `traceparent` and `tracestate`. Custom networking stacks must add the headers manually (for `x-dynatrace`, via `Dynatrace.getRequestTagHeader()` and a request tag -- see MOBL-12 §4).
3. **Backend Processing** -- The backend service (instrumented with OneAgent, OpenTelemetry, or both) reads the trace context and creates a server-side span that is linked to the mobile request.
4. **Trace Stitching** -- Dynatrace automatically stitches the mobile user action and the backend service call into a unified distributed trace.

What This Enables

Capability	Description
End-to-end latency	See total time from mobile request to backend response
Backend root cause	When TTFB is high, drill into the exact backend service, method, and database call
Service dependency mapping	Understand which backend services support which mobile features
Error correlation	Link a mobile HTTP 500 error to the specific backend exception

> **Important:** For correlation to work, the backend service must be instrumented with Dynatrace OneAgent, OpenTelemetry, or both. W3C trace context works with either.
>
> **Correction (09/28/2026).** Earlier revisions named the mobile correlation header `x-dtc`. That is the web RUM JavaScript header; the mobile SDK uses `x-dynatrace` and, from 8.333, W3C trace context. OneAgent for Mobile 8.333 was released 02/23/2026 with rollout from 02/24/2026; mobile agent versions reach users with app releases, so older builds in the field still send only `x-dynatrace`.
>
> **Sources:** OneAgent SDK for Android (DT docs) — *"To track web requests, add the x-dynatrace HTTP header with a unique value to the web request."*, What's new in OneAgent for Mobile 8.333 (DT docs) — *"OneAgent for Mobile now injects W3C Trace Context headers (traceparent and tracestate) into outbound requests"*, Frontend-backend linking (DT docs).

5. Querying Network Requests

Network requests from mobile apps are stored in the Grail `user.events` store (`characteristics.has_request`), with `url.full`, `http.request.method`, `http.response.status_code`, `duration` and `network.connection.type`. The following queries demonstrate how to retrieve, aggregate, and analyze mobile network request data.

Recent Network Requests

Retrieve the most recent network requests from mobile applications, showing the key attributes for each request.

```
// Recent network requests from mobile apps
fetch user.events, from:-1h
| filter dt.rum.application.type == "mobile" and characteristics.has_request
| fields start_time, frontend.name, url.full, http.request.method,
http.response.status_code, duration, network.connection.type
| sort start_time desc
| limit 50
```

Network Request Volume by Application

Understand which mobile applications are generating the most network traffic. This helps identify apps that may need optimization or capacity planning.

```
// Network request volume by application
fetch user.events, from:-1h
| filter dt.rum.application.type == "mobile" and characteristics.has_request
| summarize request_count = count(), by:{frontend.name}
| sort request_count desc
| limit 20
```

Trace Coverage of Mobile Requests

Measure how many mobile requests were linked to a backend distributed trace: a request event that carries `trace.id` was linked. A low rate points at backends that are not instrumented, or at app builds older than 8.333. To see the backend side, collect the `trace.id` values and query `fetch spans | filter in(trace.id, ...)`.

```
// Trace coverage of mobile-originated requests
// A request that carries trace.id was linked to backend distributed tracing
// (OneAgent for Mobile 8.333+ injects W3C traceparent / tracestate headers).
fetch user.events, from:-2h
| filter dt.rum.application.type == "mobile" and characteristics.has_request
| summarize {total_requests = count(), traced_requests =
countIf(isNotNull(trace.id))}, by:{frontend.name}
| fieldsAdd trace_rate = 100.0 * traced_requests / total_requests
| sort trace_rate asc
```

6. Slow & Failed Requests

Monitoring network request performance over time helps you detect regressions, correlate with deployments, and identify patterns. The following queries create time-series visualizations for request volume and error rates.

Request Volume Over Time

Track how network request volume changes throughout the day for each mobile application.

```
// Network request volume timeseries by application
fetch user.events, from:-24h
| filter dt.rum.application.type == "mobile" and characteristics.has_request
| makeTimeseries request_count = count(), by:{frontend.name}, interval:1h
```

Request Volume by Status Code Group

Categorize network requests by HTTP status code group (2xx, 3xx, 4xx, 5xx) to visualize error trends over time. A rising 4xx or 5xx trend may indicate API issues or backend failures.

```
// Network request volume over time by status code group
fetch user.events, from:-24h
| filter dt.rum.application.type == "mobile" and characteristics.has_request
| filter isNotNull(http.response.status_code)
| fieldsAdd status_group = if(http.response.status_code &lt; 300, then:"2xx
Success", else:if(http.response.status_code &lt; 400, then:"3xx Redirect",
else:if(http.response.status_code &lt; 500, then:"4xx Client Error", else:"5xx
Server Error")))
| makeTimeseries request_count = count(), by:{status_group}, interval:1h
```

Interpreting the Results

Pattern	What It Means	Action
Spike in 5xx errors	Backend service failures	Check backend service health, recent deployments
Rising 4xx errors	Client-side issues (bad URLs, auth failures)	Review API contracts, check token expiration logic
Request volume drop	Users abandoning the app or connectivity issues	Check app crash rates, network availability
Slow upward trend	Growing user base or increased API chattiness	Plan capacity, consider request batching

7. Performance Optimization Tips

Once you have visibility into your mobile network requests, apply these optimization strategies to improve the user experience.

Reduce Payload Sizes

Strategy	Impact
Enable gzip/brotli compression	Substantially smaller text payloads (JSON compresses well)
Use pagination	Avoid loading entire datasets at once
Return only needed fields	Use GraphQL or sparse fieldsets to minimize JSON payloads
Optimize images	Serve appropriately sized images via CDN with content negotiation

Minimize Round Trips

Strategy	Impact
Batch API calls	Combine multiple requests into a single endpoint
Use HTTP/2 or HTTP/3	Multiplexing reduces connection overhead
Implement prefetching	Load data before the user navigates to a screen
Cache responses	Use ETags and Cache-Control headers to avoid redundant requests

Leverage CDNs

Strategy	Impact
Static asset delivery	Serve images, scripts, and configs from edge locations
API acceleration	Some CDNs offer API caching and edge compute

Strategy	Impact
Geographic proximity	Reduces DNS, TCP, and TLS times for global users

Adapt to Connection Type

Strategy	Impact
Detect connection type	Use the SDK's connection type data to adjust app behavior
Low-bandwidth mode	Reduce image quality, defer non-critical requests on slow connections
Offline support	Queue requests when offline, sync when connectivity returns
Progressive loading	Load essential content first, enrich progressively

> **Tip:** Combine Dynatrace mobile monitoring data with your CDN analytics to get a complete picture of content delivery performance from edge to device.

Summary

In this notebook, we covered:

- **Automatic HTTP capture** across iOS, Android, Flutter, and React Native platforms
- **Connection type detection** and how network conditions affect request performance
- **Request timing** — the phases of an HTTP request as a diagnostic framework, and what a mobile request event actually records
- **Frontend-to-backend correlation** via the `x-dynatrace` header and, from OneAgent for Mobile 8.333, W3C trace context
- **DQL queries** on `user.events` to retrieve, aggregate, and visualize mobile network request data
- **Slow and failed request analysis** using status code grouping and time-series trends
- **Performance optimization tips** covering payload reduction, round-trip minimization, CDN usage, and adaptive behavior

Network requests are the most tangible touchpoint between your mobile app and your backend infrastructure. Monitoring them effectively ensures you can detect, diagnose, and resolve performance issues before they impact your users.

Next Steps

Continue to **MOBL-08: Session Replay for Mobile** to see how Session Replay reconstructs what the user did before a failed request or a crash.

Related Notebooks

Notebook	Topic
MOBL-06	Crash Reporting & ANR Detection
MOBL-08	Session Replay for Mobile
SPANS-01	Distributed Tracing Fundamentals

References

- Dynatrace Mobile App Monitoring
- Dynatrace Mobile SDK Documentation
- Distributed tracing (DT docs)
- Dynatrace Query Language (DT docs)

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